

Gabriel de Aragão Aleksandravicius

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Technologies

Languages: JavaScript, TypeScript, C/C++, PHP, Go, SQL, Python, HTML5/CSS3

Technologies: Node.js, ReactJS, VueJS, Next.js, Tailwind CSS, PostgreSQL, MongoDB, LangChain, Redis, WebSockets, MQTT, Express.js, Matter

Tools: Linux, Docker, Git, Keycloak, Gitlab CI, GitHub Actions, Altium Designer

Practices: Domain-Driven Design (DDD), Test-Driven Development, CI/CD, Agile, REST API Design, Microservices, Event-Driven Architecture

Experience

Sentosa Technology Consultants (contractor)

April 2024 – Present

Full Stack Software Engineer / Embedded Software Engineer

Remote (Bern, CH)

- **Cablelabs - Gateway-as-a-Platform**

- Built core platform components for an edge/IoT gateway (Matter, Docker/OCI, embedded Linux) and developed full-stack smart-home automation app (containerized device app + cloud/SPA backend)
- Integrated multi-service architecture across app store, cloud, and hardware layers and validated complete system on physical hardware for CableLabs Tech Expo 2026 demo

- **Cablelabs - Quality By Design / Network Side Probing**

- Embedded software engineer working on a research project that aims to guarantee network quality for end users by detecting degradation and automatically triggering network-level interventions.
- Developing the measurement layer directly on the home router: detecting which application servers a user is communicating with and passively measuring connection quality.

- **Cablelabs - STAMP Protocol Latency Testing Platform:** Designed and built a web app and embedded services using socket programming for STAMP packet generation, transmission, and precise timing.

- **SelfAudit AI - CMMC Compliance Automation Platform (AI-Powered):** Designed and built a RESTful API (Node.js/TypeScript), PostgreSQL schema, and Keycloak/CASL auth, with secure webhook/REST integration to an AI engine.

CERN

Jan 2021 – Mar 2024

Full Stack Software Developer

Geneva, CH

- Developed and maintained Glance, the ATLAS collaboration's management system for personnel, speaker selection, and publication workflows.
- Built REST APIs in PHP with Domain-Driven Design to support VueJS frontends; modernized legacy code through incremental refactoring.
- Led integration of ATLAS member registration with CERN HR API, automating workflows and saving up to 1 hour daily for the Secretariat.
- Automated dynamic synchronization between Glance and CERN HR databases, reducing manual work by 30 minutes daily.
- Dockerized Glance database for consistent local development; introduced ephemeral databases and Liquibase migrations in GitLab CI/CD to enhance testing and deployment.

UFRJ Aerospace Research Group

Sep 2019 – Dec 2021

Avionics Engineer / Manager

Rio de Janeiro, Brazil

- Led a team of 4 in developing a PC/104 flight computer for data acquisition, telemetry, and parachute deployment, using Altium Designer for hardware and embedded C/C++ for software.
- Deployed avionics system in the 2023 Latin American Space Challenge: 1st place in the 3 km Solid Motors category; 2nd place in 2020 in both Solid Motors and Hybrid/Liquid categories.

Education

Haute École Arc (HE-Arc) - Neuchâtel

Sep 2025 (ongoing)

- Bachelor of Science in Computer Science, Major in Embedded Software

University of Rio de Janeiro

Mar 2018 – Dec 2021 (discontinued)

- Bachelor of Science in Bionanotechnology

Languages

English/Portuguese (C2), French (C1), German (A1)

Publications

Enhancing data consistency in ATLAS and CERN HR databases through automated synchronization

May 2023

Gabriel de Aragão Aleksandravicius et al.

<https://doi.org/10.1051/epjconf/202429505004>

Additional Information

- **CHEP 2023** (Norfolk, USA): Gave a [talk](#) at the 26th International Conference on Computing in High Energy Nuclear Physics on automated data synchronization between ATLAS Glance and CERN HR databases.
- **ATLAS Speakers Committee Workshop** (CERN, 2022): Presented ongoing and planned developments in ATLAS Glance tooling.